

HITESH KUMAR PUNNA

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SUMMARY

Quant researcher with experience in systematic options strategies and ML-driven signal tools. Pursuing a Master's in Financial Engineering and seeking roles blending machine learning and quantitative finance.

EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY, Scheller College of Business

Atlanta, GA

Master of Science in Quantitative and Computational Finance

Aug 2025 – May 2027

- GPA: N/A (joining Fall 2025)
- Key Coursework: C++, Stochastic Calculus, Derivatives, Statistical Learning, Fixed Income, Numerical Methods

INDIAN INSTITUTE OF TECHNOLOGY BOMBAY

Mumbai, India

Bachelor of Technology in Computer Science and Engineering

Jul 2019 – May 2023

- GPA: 8.86/10
- Minor in Applied Statistics and Honors in Computer Science

EXPERIENCE

GRAVITON RESEARCH CAPITAL LLP

Delhi, India

Quant Researcher and Analyst

Jun 2023 – Jul 2025

- Built and deployed a high-performing **stock options strategy** on the NSE, managing the full pipeline from signal research to live execution and transitioning it into production with consistent returns.
- Scaled an existing commodity options strategy on **MCX**, expanding from 2 to 6 instruments including **GOLD, SILVER**, and their Mini contracts, resulting in a **2x increase in strategy profitability**.
- Developed **XGBoost classifiers** trained on 50+ features (liquidity, Greeks, volatility) to denoise signals, identify market regimes, and optimize entries resulting in a **10–20% gain in net PnL**.
- Automated hyperparameter tuning using **Genetic Algorithms**, reducing model training time from **120 minutes to under 15 minutes** and accelerating the research-to-deployment cycle.
- Modeled complex derivatives such as **Callable Bull/Bear Contracts (CBBs)** using multi-factor **linear regression**, improving pricing accuracy and risk control in low-liquidity environments.

GRAVITON RESEARCH CAPITAL LLP

Delhi, India

Quant Research Intern

May 2022 – Jul 2022

- Designed microstructure based market indicators using statistical filters to capture short-term trend strength and intraday regime shifts, aiding signal generation in high-frequency environments.
- Trained and deployed ML models to extend traditional linear regression frameworks using existing alphas and few market indicators; boosted NSE Index Options strategy PnL by **10%** in backtesting.
- Enhanced alpha performance for both stock and index options books by tuning weighting functions and suppressing poor regimes, improving returns from **6% to 8%** and **8% to 10%**, respectively.

EXO FLY (Tech Team IIT Bombay)

Mumbai, India

Student Team Lead

Mar 2022 – Apr 2023

- Led a **6-member team** and secured funding to design a safe and compact manned **eVTOL aerial vehicle**
- Designed a controller on **Simulink**, performing **sensor fusion** for localization with an **Extended Kalman Filter**

PROJECTS

Optimization of Deep Neural Networks via Layer Pruning

Jan - April 2023

- Employed **layer and node pruning**, and training **student-teacher networks**, to optimize the size of neural networks.
- Achieved a significant one-thirds reduction in model size while ensuring accuracy remained within a 5% margin.

MDP Optimization with Reinforcement Learning

Sep - Dec 2021

- Explored sampling algorithms like **KL-UCB** and **Thompson Sampling** for stochastic multi-armed bandits
- Programmed **SARSA** with linear approximation for the mountain car game using **OpenAI Gym** in Python

SKILLS

Programming Languages	C++, Python, R, Bash, PyTorch, TensorFlow, scikit-learn, OpenCV, MATLAB
Interest	Badminton, Rock Climbing, Competitive Coding